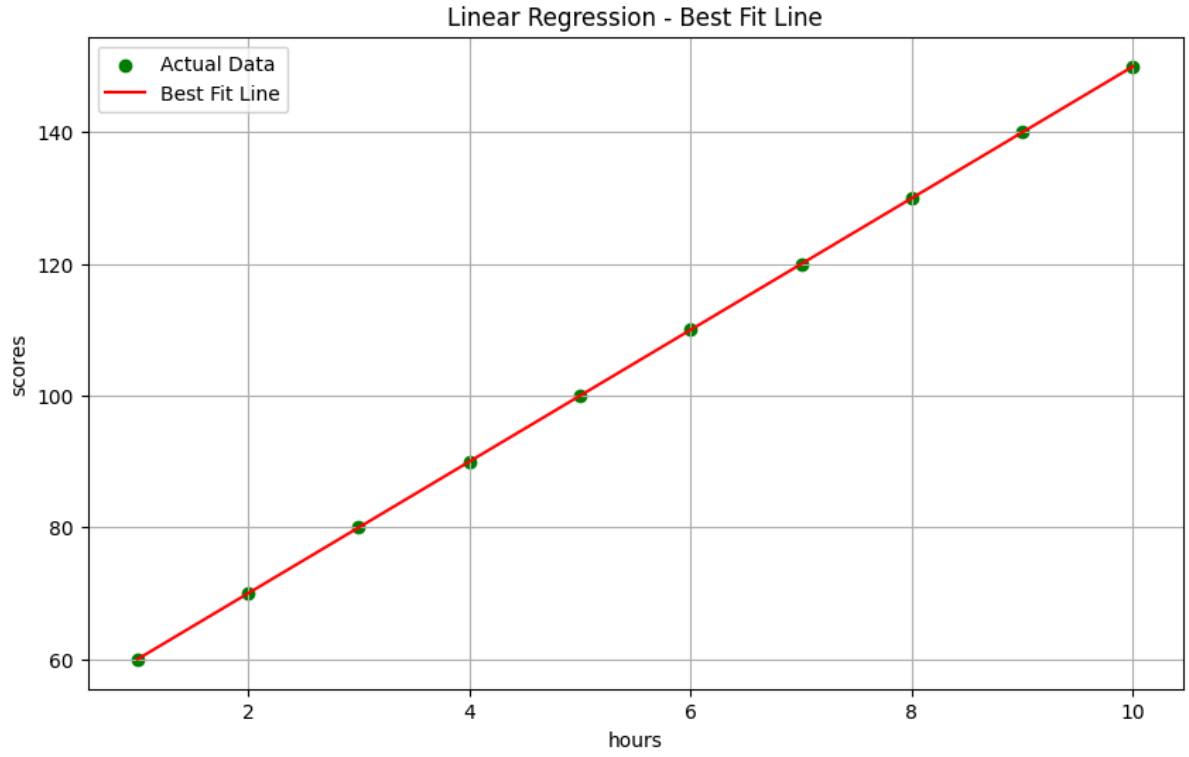
```
import numpy as np
import matplotlib.pyplot as plt
```

```
hours=np.array([1,2,3,4,5,6,7,8,9,10])
scores=np.array([60,70,80,90,100,110,120,130,140,150])
```

```
predicted_scores = 10 * hours + 50
plt.figure(figsize=(10,6))
plt.scatter(hours, scores, label="Actual Data", color="green")
plt.plot(hours, predicted_scores, label="Best Fit Line", color="red")
plt.xlabel("hours")
plt.ylabel("scores")
plt.title("Linear Regression - Best Fit Line")
plt.legend()
plt.grid(True)
plt.show()
```




```
from matplotlib.cbook import delete_masked_points
x=np.array([1,2,3,4,5],dtype=float)
y=np.array([60,70,80,90,100],dtype=float)
m=0
c=0
learning_rate=0.01
epochs = 1000
n=len(x)
for i in range(epochs):
  y_pred=m*x+c
  error=y-y_pred
  mse=np.mean(error ** 2)
  dm=(-2/n)*np.sum(x* error)
```

```
  dc=(-2/n)*np.sum(error)
  m=m-learning_rate*dm
  c=c-learning_rate*dc
  if i % 100 == 0:
    print(f"Epoch {i}")
    print(f"MSE = {mse:.2f}")
    print(f"m = {m:.2f},c={c:.2f}")
    print("----------------------")
    print("\nfinal best fit line:")
    print(f"y={m:.2f}x={c:.2f}")
```

```
final best fit line:
y=0.00x=0.00
Epoch 200
MSE = 0.00
m = 0.00,c=0.00
----------------------

final best fit line:
y=0.00x=0.00
Epoch 300
MSE = 0.00
m = 0.00,c=0.00
----------------------

final best fit line:
y=0.00x=0.00
Epoch 400
MSE = 0.00
m = 0.00,c=0.00
----------------------

final best fit line:
y=0.00x=0.00
Epoch 500
MSE = 0.00
m = 0.00,c=0.00
----------------------

final best fit line:
y=0.00x=0.00
Epoch 600
MSE = 0.00
m = 0.00,c=0.00
----------------------

final best fit line:
y=0.00x=0.00
```

```
final best fit line:
y=0.00x=0.00
Epoch 900
MSE = 0.00
m = 0.00,c=0.00
----------------------

final best fit line:
y=0.00x=0.00
```

```
from sklearn.model_selection import train_test_split
```

```
experience=np.array([1,2,3,4,5,6,7,8,9,10,11,12,13,14,15])
salary=np.array([35,38,42,48,55,60,65,70,76,82,87,92,96,100,105])
x=experience.reshape(-1,1)
y=salary
x_train,x_test,y_train,y_test=train_test_split(
    x,y,test_size=0.2,random_state=42)
print(f"Training samples:{len(x_train)}")
print(f"Testing samples:{len(x_test)}")
```

```
Training samples:12
Testing samples:3
```

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
print(f"slope(coefficent):{model.coef_[0]:.2f}")
print(f"intercept:{model.intercept_:.2f}")
```

```
slope(coefficent):5.22
intercept:27.94
```

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
print(f"slope(coefficent):{model.coef_[0]:.2f}")
print(f"intercept:{model.intercept_:.2f}")
y_pred=model.predict(x_test)
for actual,predicted in zip(y_test,y_pred):
  print(f"Actual:{actual:>5},Predicted:{predicted:>7.1f}")
  (f"error: {predicted - actual:+.1f}")
```

```
slope(coefficent):5.22
intercept:27.94
Actual:   82,Predicted:   80.2
Actual:   92,Predicted:   90.6
Actual:   35,Predicted:   33.2
```

```
from sklearn.metrics import r2_score,mean_absolute_error,mean_squared_error
import numpy as np
```

```
y_actual = np.array([500, 300, 700, 450, 600])
y_pred = np.array([480, 320, 690, 460, 590])
r2 = r2_score(y_actual, y_pred)
mae = mean_absolute_error(y_actual, y_pred)
```

```
rmse = np.sqrt(mean_squared_error(y_actual, y_pred))
print(f"R² Score : {r2:.4f} → model explains {r2*100:.1f}% of variance")
print(f"MAE : {mae:.1f} → avg error ± {mae:.0f} units")
print(f"RMSE : {rmse:.1f} → penalised avg error {rmse:.0f} units")
```

```
R² Score : 0.9880 → model explains 98.8% of variance
MAE : 14.0 → avg error ± 14 units
RMSE : 14.8 → penalised avg error 15 units
```

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error
np.random.seed(42)
n = 100
data = pd.DataFrame({
 'experience': np.random.randint(0, 20, n),
 'education': np.random.randint(12, 20, n),
 'age': np.random.randint(22, 55, n),})
data['salary'] = (
    data['experience'] * 4500 +
    data['education'] * 2000 +
    data['age'] * 300 +
    np.random.normal(0, 3000, n) +
    10000)
X = data[['experience', 'education', 'age']]
y = data['salary']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
for feat, coef in zip(X.columns, model.coef_):
 print(f" {feat:12s}: ${coef:,.0f} per unit increase")
print(f" Base salary (intercept): ${model.intercept_:,.0f}")
y_pred = model.predict(X_test)
print(f"\nR² : {r2_score(y_test, y_pred):.3f}")
print(f"MAE : ${mean_absolute_error(y_test, y_pred):,.0f}")
```

```
 experience  : $4,534 per unit increase
 education   : $2,203 per unit increase
 age         : $266 per unit increase
 Base salary (intercept): $7,694

R² : 0.988
MAE : $2,219
```

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
np.random.seed(42)
n = 200
locations = ['Urban', 'Suburban', 'Rural']
df = pd.DataFrame({
 'area_sqft': np.random.randint(500, 4000, n),
 'num_rooms': np.random.randint(1, 7, n),
```

```
 'location': np.random.choice(locations, n),
 'age_years': np.random.randint(0, 40, n),
})
loc_map = {'Urban': 1.5, 'Suburban': 1.0, 'Rural': 0.7}#expensiv medium cheap
noise = np.random.normal(0, 20000, n)
df['price'] = (
 df['area_sqft'] * 150 +
 df['num_rooms'] * 25000 -
 df['age_years'] * 2000 +
 df['location'].map(loc_map) * 50000 +
 noise
).astype(int)
print(df.head(8).to_string(index=False))
print(f"\nDataset shape : {df.shape}")
print(df.describe()[['area_sqft','num_rooms','age_years','price']].round(0))
```

```
 area_sqft  num_rooms location  age_years  price
      3674          4    Rural         20 646067
      1360          4    Urban         35 300479
      1794          6    Urban          9 501609
      1630          6    Rural         36 355668
      1595          3    Urban          8 391991
      3592          2    Rural         23 566413
      2138          4 Suburban         34 410974
      2669          1    Urban         34 459191

Dataset shape : (200, 5)
       area_sqft  num_rooms  age_years     price
count      200.0      200.0      200.0     200.0
mean      2334.0        3.0       20.0  451548.0
std        992.0        2.0       11.0  154871.0
min        521.0        1.0        0.0   86373.0
25%       1519.0        2.0       10.0  326188.0
50%       2372.0        3.0       20.0  458282.0
75%       3230.0        5.0       30.0  575910.0
max       3999.0        6.0       39.0  781261.0
```

```
le=LabelEncoder()
df['location_enc'] = le.fit_transform(df['location'])
features = ['area_sqft', 'num_rooms', 'age_years', 'location_enc']
X = df[features]
y = df['price']
print(df[['location', 'location_enc']].drop_duplicates().sort_values('location_enc')
print(f"\nFeature matrix shape : {X.shape}")
print(f"Target vector shape : {y.shape}")
```

```
   location  location_enc
0     Rural             0
6  Suburban             1
1     Urban             2

Feature matrix shape : (200, 4)
Target vector shape : (200,)
```

```
X_train, X_test, y_train, y_test = train_test_split(
 X, y, test_size=0.2, random_state=42
```

```
)
print(f"Training samples : {len(X_train)} ({len(X_train)/len(X)*100:.0f}%)")
print(f"Testing samples : {len(X_test)} ({len(X_test)/len(X)*100:.0f}%)")
print(f"Feature names : {features}")
```

```
Training samples : 160 (80%)
Testing samples : 40 (20%)
Feature names : ['area_sqft', 'num_rooms', 'age_years', 'location_enc']
```

```
model = LinearRegression()
model.fit(X_train, y_train)
print("Learned Coefficients:")
print("-" * 35)
for feat, coef in zip(features, model.coef_):
 direction = "↑" if coef > 0 else "↓"
 print(f" {feat:13s}: Rs {coef:>10,.0f} {direction}")
print(f" {'intercept':13s}: Rs {model.intercept_:>10,.0f}")
print("-" * 35)
```

```
Learned Coefficients:
-----------------------------------
 area_sqft    : Rs        151 ↑
 num_rooms    : Rs     23,169 ↑
 age_years    : Rs     -2,262 ↓
 location_enc : Rs     18,114 ↑
 intercept    : Rs     46,789
-----------------------------------
```

```
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
print("=" * 50)
print(" MODEL EVALUATION REPORT")
print("=" * 50)
print(f"R² Score : {r2:.4f} {
    'Excellent' if r2>0.9 else 'Good' if r2>0.8 else ' Needs work'}")
print(f"MAE : Rs {mae:>10,.0f}")
print(f"RMSE : Rs {rmse:>10,.0f}")
print("=" * 50)
print("\nSample Predictions:")
print(f"{'Actual':>10} | {'Predicted':>10} | {'Error':>10}")
print("-" * 46)
for a, p in zip(y_test[:8], y_pred[:8]):
 print(f"{a:>10,} | {p:>10,.0f} | {p-a:>+10,.0f}")
```

```
==================================================
 MODEL EVALUATION REPORT
==================================================
R² Score : 0.9872 Excellent
MAE : Rs     13,439
RMSE : Rs     16,959
==================================================

Sample Predictions:
    Actual |  Predicted |      Error
----------------------------------------------
   489,899 |    478,320 |    -11,579
```

```
   601,329 |    572,877 |    -28,452
   623,788 |    650,489 |    +26,701
   291,458 |    271,086 |    -20,372
   390,040 |    344,431 |    -45,609
   123,884 |    134,545 |    +10,661
   484,442 |    443,295 |    -41,147
   562,823 |    556,350 |     -6,473
```

```
new_houses = pd.DataFrame({
 'area_sqft': [230, 250,350, 500],
 'num_rooms': [2, 5, 1, 6 ],
 'age_years': [7, 10, 25, 2 ],
 'location_enc':[2, 1, 0, 2 ],
 'location_lbl':['Urban','Suburban','Rural','Urban'],
})
predictions = model.predict(new_houses[features])
print("\n HOUSE PRICE PREDICTIONS")
print("=" * 60)
for i, (_, row) in enumerate(new_houses.iterrows()):
 print(f" House {i+1}: {row.area_sqft} sqft | "
 f"{row.num_rooms} rooms | "
 f"{row.age_years}yr old | "
 f"{row.location_lbl}")
 print(f" Estimated Price → Rs {predictions[i]:>10,.0f}")
 print()
```

```
 HOUSE PRICE PREDICTIONS
============================================================
 House 1: 230 sqft | 2 rooms | 7yr old | Urban
 Estimated Price → Rs    148,154

 House 2: 250 sqft | 5 rooms | 10yr old | Suburban
 Estimated Price → Rs    195,771

 House 3: 350 sqft | 1 rooms | 25yr old | Rural
 Estimated Price → Rs     66,109

 House 4: 500 sqft | 6 rooms | 2yr old | Urban
 Estimated Price → Rs    292,797
```